A. Seniors / Older Adults — Question Bank + Model Answers

Q1) You designed a medicine reminder app for seniors. Which evaluation method(s) will you use and why?

Model Answer (write like this):
I would run **formative usability testing with task-based scenarios** because seniors often face higher error risk from stress, memory load, or unfamiliar interaction patterns, so we must observe *real task performance* early and iterate . I'll recruit seniors matching the persona (age range, smartphone experience, vision/motor limits) because "**know your user**" matters: designers are not a good proxy .
During testing, I'll give realistic tasks: "set a reminder," "mark dose as taken," "edit time," "recover from a wrong tap." I'll measure **task completion rate, time-on-task, error rate, help requests, and critical errors**, and I'll watch for **slips vs mistakes**—slips show interface execution problems; mistakes show a wrong mental model and need better feedback and mapping . Then I'll iterate (prototype → test → redesign) because the process is not one-shot .

Q2) A senior user says: "I don't know what to do next" on your checkout flow. What evaluation technique best reveals the issue?

Model Answer:
This is best revealed through **moderated usability testing + observation**, because the goal is to detect breakdown points in the flow (where they hesitate, backtrack, or ask questions). That maps directly to navigation clarity: a good system should help the user know **where they are**, **what they can do**, **where they are going**, and **where they've been** .
I would run tasks like "add item → checkout → confirm payment," and note where the user loses orientation. If they can't infer the next step, we redesign using stronger signposting (progress indicator, clearer labels, feedback after each step). Then re-test to verify improvement (iterative design loop) .

Q3) For seniors, what should you change in *how* you run the evaluation session?

Model Answer:
I would adapt the protocol to reduce anxiety and cognitive load because stress increases difficulty even for simple tasks (negative affect narrows thinking). So I'll use **short task blocks**, clear instructions, and allow breaks. I'll avoid rushing and I'll phrase tasks in everyday language (not UI labels).
I'll also use **observation + follow-up interview**: observation tells me what happens; interview tells me *why* (what mental model they formed). This aligns with user-centered research methods

like **talking to users (interviews)** and **watching users (observation)** in requirements/evaluation cycles .

Q4) You want to compare "Design A vs Design B" for seniors. Which study type is appropriate?

Model Answer:
I would use an **experimental comparison** (A/B style) if my aim is cause-and-effect: does Design A *cause* fewer errors or faster completion than Design B. That is "experimental research" where we manipulate an **independent variable** (design version) and measure a **dependent variable** (time, errors, success) .
To make results trustworthy, I would randomize order (or use counterbalancing) because seniors may learn the interface on the first attempt. If true experimentation is too heavy, I can still do **descriptive testing** to capture where seniors struggle, but then I must not claim causation .

B. Disabled Users — Question Bank + Model Answers (sensory + motor + cognitive)

Q5) Your website must work for screen reader users. How do you evaluate it properly?

Model Answer:
I would use a mix of **expert accessibility inspection** + **assistive-technology user testing**. First, do heuristic-style checks focused on perceivability/operability (structure, focus order, meaningful labels). Then validate with real screen reader users performing tasks, because expert review alone can miss real navigation pain points.
In the user test, tasks should include: "navigate headings," "find and activate primary CTA," "fill form," "recover from an error." I'll measure task completion, time, and critical failures, and I'll watch for navigation breakdown against the "four navigation rules" (knowing where you are / what you can do / where you're going / where you've been) . Findings feed into redesign and re-test (iterative evaluation loop) .

Q6) Your app targets users with motor impairments (shaky hands). Which evaluation tasks/metrics matter most?

Model Answer:
I would run **task-based usability testing** with emphasis on error patterns that are predictable under reduced physical skill: missed taps, accidental double taps, wrong target selection. Those are often **slips** (right intention, wrong execution), so the design fix is usually larger touch targets, spacing, and forgiving interaction (undo/confirm for destructive actions) .
Metrics I'd prioritize: **mis-tap rate**, number of retries, time to complete, and whether the user

can recover without help. I'd also include scenarios that test error recovery because designing for human error is core to HCI's "to err is human" mindset: errors are normal and systems should be resilient .

Q7) How do you evaluate a design for users with color-vision deficiency?

Model Answer:
I would do **visual inspection and variant testing** where information is interpreted without color: check whether status, categories, and alerts are still distinguishable using shape, text, icons, or position. Then I'd do user testing (or at minimum structured review) focusing on tasks like "identify warnings," "read chart legends," "choose selected item."
The core evaluation question is: does the interface remain understandable if color is removed? If not, redesign using redundant coding (icon + label + pattern). This reduces perception-based confusion and avoids mistakes caused by wrong interpretation of system state (evaluation gap) .

Q8) For cognitive disabilities (attention/memory limits), what evaluation method best exposes problems?

Model Answer:
I would use **formative usability testing with observation** and focus on cognitive load: where users forget steps, lose context, or can't recall options. Then I'd explicitly check whether the UI relies on **recall** instead of **recognition** (recognition is easier and more reliable).
I would measure: number of times they re-open menus, backtrack, ask "what now?", or abandon tasks. Design-wise, I would reduce memory load using visible state, clear feedback, and consistent structure, because the goal is to bridge the gulf: make actions obvious and show clearly what happened .

C. Illiterate / Low-Literacy Users — Question Bank + Model Answers

Q9) You built a government service kiosk for low-literacy users. How will you evaluate it?

Model Answer:
I would prioritize **field-based usability testing (contextual testing)** because literacy issues are strongly linked to real context: stress, queues, noise, fear of making mistakes. Lab-only testing can hide those constraints. I'd recruit actual low-literacy participants and run "do-this" tasks using spoken prompts instead of written instructions, while observing where they hesitate or ask for help (user-centered observation) .
Metrics: task completion, wrong-path rate, reliance on assistance, and confusion around icons. I

would also do short **post-task interviews** using simple oral questions (“What did you think this button means?”) to see if icon mental models match intent. Then redesign with stronger signifiers, clearer grouping, and reduced text reliance, and iterate again .

Q10) What evaluation technique helps you test whether your icons are understood by illiterate users?

Model Answer:
I would use a **comprehension test + observation**: show the icon in context and ask the user what action they expect (oral, not written). Then confirm by giving a task where they must choose the correct icon under time pressure (realistic).
This works because low-literacy users may not read labels, so meaning must come from recognition and clear signifiers. If many users misinterpret, it’s a “mistake” issue (wrong intention due to wrong mental model), and redesign must clarify mapping and feedback rather than just making the icon bigger .

Q11) Why is a standard written questionnaire a weak evaluation method for illiterate users, and what do you use instead?

Model Answer:
A written questionnaire fails because it assumes reading fluency and can produce fake agreement (participants guess answers or say “yes” to finish quickly). That creates invalid data. Instead, I’d use **structured interviews**, **observation**, and very short **oral rating scales** (e.g., “easy / ok / hard” with emoji faces) to capture satisfaction.
The strongest evidence still comes from task performance: whether the user can complete key tasks independently. That aligns with evaluation as “assessing the prototype against requirements” using user testing and feeding results into redesign .

D. Teenagers — Question Bank + Model Answers

Q12) You designed a mental-wellbeing app for teens. What evaluation approach is most appropriate?

Model Answer:
I would use a **mixed-method approach**: early **formative usability testing** to remove friction (can they complete onboarding, set goals, find help tools), plus **longitudinal/diary-style feedback** (or repeated sessions) to capture engagement over time. Teens’ needs are strongly influenced by motivation, social context, and daily routines, so one-off testing can miss drop-off reasons.
I’d measure: time to first success, feature discoverability, and reasons for abandonment (via short interviews). Because teens care about autonomy and trust, I’d explicitly evaluate privacy

clarity: do they understand what data is collected and what is shared? Confusion here creates negative affect and reduces continued use .

Q13) A teen user says: “This feels cringe / boring.” How do you evaluate and respond, in HCI terms?

Model Answer:
I would treat this as a usability + experience issue: not only “can they use it,” but “do they want to use it.” I’d run **semi-structured interviews** after tasks to identify which parts create negative emotion and why (tone, visuals, microcopy, too many steps). Since emotion affects cognition, negative affect narrows thinking and makes tasks feel harder, so improving tone and reducing friction can improve performance too .
Then I’d iterate prototypes and re-test. If we’re comparing variants of tone or UI style, I can frame it as an experiment (Design A vs B) and measure retention or completion differences while being careful about causation claims .

Q14) Which evaluation method helps you test teen onboarding where the goal is “minimum confusion”?

Model Answer:
I would run a **first-time user experience usability test**: give them the app with minimal instruction and observe whether they can form the right mental model quickly. I’d watch the exact moments they ask “what can I do?” or “what happened?”, which maps to the **gulfs of execution and evaluation**: make actions obvious and feedback clear .
Metrics: time to complete onboarding, number of backtracks, and the number of times they open help. Then fix by improving signifiers, simplifying steps, and showing immediate value early.

E. “Sir-Style” Combo Questions (very likely in exam)

Q15) For each group (seniors, disabled, illiterate, teens), pick ONE best evaluation method and justify it.

Model Answer (structured):

- **Seniors: Moderated task-based usability testing**. Seniors vary widely in experience; observation reveals where navigation, font, and error recovery fail. Measure completion/time/errors and redesign iteratively .
- **Disabled users: Accessibility-focused testing with assistive tech + expert review.** Real AT usage reveals issues experts miss; evaluate focus order, labels, keyboard navigation, and critical task success.

- **Illiterate/low literacy: Contextual/field usability testing + oral interview.** Written surveys fail; direct observation + spoken prompts show icon comprehension and independence.
- **Teens: Mixed method: usability + longitudinal engagement feedback.** Teens' adoption depends on motivation and emotion; repeated feedback catches drop-off reasons and trust issues .

Q16) A teacher asks: "Which study type is this: 'Does age relate to error rate?'"

Model Answer:
That is a **relational investigation** because it seeks a relationship/correlation between age and errors but does not manipulate variables. It can show that variables move together, but it usually cannot claim causation. If we want causation, we need an experimental design with controlled manipulation .